

CS 6343 HOMEWORK #1

SPRING 2026

Question 1 (15 points):

AES Avalanche, ShiftRows, and MixColumns

Scenario: DES is being phased out, and NIST is running a public competition to select a new encryption standard. Multiple teams have submitted candidate block ciphers, and Rijndael (the future AES) is one of the finalists. You work in a research lab tasked with evaluating candidates by inspecting their round structure and investigating dynamics of avalanche/diffusion, and whether specific design choices are doing their intended jobs. In this homework, you will “look inside” AES by recording intermediate round states and running controlled experiments (e.g., modifying ShiftRows/MixColumns) to assess how these components contribute to diffusion and overall security intuition.

FYI: Your AES code is a toy tracing implementation meant to study bit/byte dynamics. It does not include many protections required for production use (e.g., constant-time behavior, side-channel hardening, misuse resistance), so it should not be used for real security.

Tools allowed

- You may use an AES library **only** for baseline encryption correctness.
- For internal states, implement the round steps yourself (SubBytes, ShiftRows, MixColumns, AddRoundKey) and log intermediate states.
- $GF(2^8)$ operations: use AES `xtime()` (mul-by-2) and derive mul-by-3, or use a GF library.

Experiment inputs

Use AES-128 with:

- Key $K = 00\ 00\ \dots\ 00$ (16 bytes)
- Plaintext $P = 00\ 00\ \dots\ 00$
- Plaintext P' : flip 1 bit of P (e.g., set byte0 to 01)

Part A: Avalanche across rounds (plaintext bit flip)

Implement the AES algorithm as presented in class. Run AES on P and P' and record, for rounds **0 through 10**:

- state after AddRoundKey (round 0)

- state after each round's SubBytes, ShiftRows, MixColumns, AddRoundKey. Note that round 10 omits MixColumns.

For each stage compute:

- **Byte Hamming distance**
- **Bit Hamming distance**
 1. By which round is diffusion close to half of the bits being flipped?
Describe what you see.
 2. Repeat the above experiment with 10 different plaintexts having a single bit flipped in different positions relative to the original all zeros plaintext and discuss the general observed patterns.

Part B: Disable ShiftRows

Modify AES so ShiftRows does nothing, keep everything else unchanged. Repeat Part A (rounds 0–10).

Part C: Modify MixColumns

Replace the matrix used in MixColumns with:

$$M_{\text{new}} = \begin{bmatrix} 01 & 01 & 00 & 00 \\ 00 & 01 & 01 & 00 \\ 00 & 00 & 01 & 01 \\ 00 & 00 & 00 & 01 \end{bmatrix}$$

Keep ShiftRows enabled. Repeat Part A (rounds 0–10). Discuss the impact of the matrix change to the AES diffusion mechanism.

Part D: Key-bit flip experiment

Fix plaintext $P = 0^{128}$. Compare keys:

- K : all zeros
- K' : flip 1 bit (e.g., first byte 01)

Repeat Part A, B and C for rounds 0–10 for K vs K' (This time you leave the plaintext unchanged but only make changes to the key).

Tabulate and discuss your observations.

Question 2 (7 points):**Modes of Operation**

In this activity, you will do some hands-on work with the modes of operation that we studied in class. You will provide descriptions of your observations at each stage, supported by screen shots showing what you observed.

You will use OpenSSL, a program and library that supports a wide range of cryptographic operations. My personal choice for this kind of work has tended to be Kali linux since Kali comes with lots of other cool security things to play about with (being a windows user, this means a virtualbox Kali VM on the windows system).

If my own choice resonates with you, you may find a prebuilt Kali VM here:
<https://www.osboxes.org/kali-linux/#kali-linux-2017-03-vbox>

... and of course virtualbox itself here:

<https://www.virtualbox.org/wiki/Downloads>

You are free to do the project in an environment of your choice. You don't have to follow my choices.

In this exercise you will compare the behavior of the CBC and ECB modes of operation. You will use the provided image file, Secret.bmp, for these comparisons.

- (a) Before doing anything with the image file, view it using an image viewer of your choice. This will give you a baseline for the judgments you will make post encryption.
- (b) Create a random 16-byte key and 16-byte IV that you will use for AES operations.

Encrypt the image file provided using AES in ECB mode and in CBC mode. You now have two different encrypted files. Try to view these two encrypted files using the image editor you used previously. What do you see and why?

- (c) To correct the issues you should have met above, you will use the help of a hex editor. Install a hex editor of your choice and use it to open the original unencrypted image file. Now that you can directly view the bytes in the file, copy the first 54 bytes of Secret.bmp and paste them to replace the first 54 bytes of each of the encrypted files.

Use the image viewer to view each of the two new files you have created. What explains the difference in behavior to what you observed in (b)?

Explain/Justify the difference in observations between the case when you used the ecb options and when you used the cbc options.

Question 3 (8 points):

TLS Cipher Suites

In this activity, you will observe TLS traffic between your client and a server in order to make connections with cryptographic algorithms learned in class. As part of your submission, you will include screenshots to support your wireshark traffic observations.

- a) Install wireshark on your computer.
- b) With wireshark running, use your browser to browse a page from a TLS-enabled website of your choice (i.e., a website supporting https; which is pretty much every serious website today).
- c) Observe packets from the TCP 3-way handshake of your connection. You may click on any packet in this connection and select the option to follow the stream to reduce the clutter and see only the packets from this connection.
- d) Select the *TLS Client Hello* following the 3-way handshake of your connection. In the bottom-left corner of your Wireshark window, expand the relevant section and observe the cipher suites proposed by your browser. For each proposed cipher suite, identify the following 5 classes of algorithms when applicable: (1) Key Exchange Algorithm; (2) Authentication Algorithm; (3) Cipher/Encryption Algorithm; (4) Cipher Mode; (5) MAC Algorithm. If you encounter a TLS 1.3 cipher suite, briefly explain why this 5-part breakdown does not map exactly the same way as in TLS 1.2, and describe the corresponding cryptographic components as accurately as possible.
- e) Go back to the main window and select the *TLS Server Hello* for this connection. Observe the cipher suite ultimately selected by the server for your connection
- f) Repeat (b) through (e) for two different browsers and eight more TLS-enabled websites.
- g) In your small sample of websites, what are the most commonly used algorithms in the server and browser cipher suites? What might be the justifications behind these choices?